Determine tight PTAS running times for q-norm load balancing with intermediate machines

`W4413848299_p0` (website page)

Website #26 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

Fix a constant $q > 1$. For an instance with m identical machines and jobs $j = 1, \dots, n$ of processing times $p_j > 0$, a schedule induces loads

$$C_i = \sum_{j \to i} p_j \qquad (i = 1, \dots, m),$$

and objective

$$F(\sigma) := \sum_{i=1}^{m} C_i^q.$$

The optimization problem is $P \| \| \sum_i C_i^q$.

The user asks for the tight asymptotic running time, as a function of ε and n, of the fastest PTAS when

$$m = \Theta\big((1/\varepsilon)^\theta\big), \qquad \theta \in (1/2, 1].$$

Equivalently: either prove a matching ETH lower bound in this regime, or design a faster PTAS.

What I will actually prove is the following rigorous partial result: for every fixed $\theta \in (1/2, 1)$, any PTAS in the regime $m = \Theta((1/\varepsilon)^\theta)$ must, under ETH, take subexponential but superpolynomial time

$$2^{\Omega((1/\varepsilon)^{1-\theta-o(1)})} + n^{O(1)}.$$

This does not close the problem, but it is a genuine new lower bound throughout the whole open interval $(1/2, 1)$.

Alternative readings considered

The strongest reading is: completely characterize the optimal PTAS dependence on $1/\varepsilon$ for every fixed $\theta \in (1/2, 1]$, with matching upper and lower bounds. I adopt that as the target reading.

A weaker reading is: make rigorous progress by proving either a new ETH lower bound or a faster PTAS in some nontrivial subregime. My result falls under this weaker reading.

The endpoint $\theta = 1$ is part of the original question; my theorem is nontrivial only for $\theta \in (1/2, 1)$. I state this limitation explicitly below.

Related literature found

- **Chen–Tao–Verschae (SODA 2022 / MOR 2025), classification: PARTIAL.** They prove a PTAS with running time $2^{\tilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}$, an ETH lower bound excluding $2^{O((1/\varepsilon)^{1/2-\delta})} + n^{O(1)}$ for any $\delta > 0$, and they explicitly note that their lower bound applies when $m = \Theta(\sqrt{1/\varepsilon})$ while the regime $m = \Theta((1/\varepsilon)^\theta)$, $\theta \in (1/2, 1]$, remains open. This is the main source I use as a tool.

- **Ibrahimpur–Swamy (ICALP 2021), classification: BACKGROUND.** They prove a PTAS for minimum-norm load balancing on identical machines for arbitrary monotone symmetric norms, so in particular for ℓ_q-norms. This confirms that the issue here is not existence of a PTAS but fine-grained dependence on ε and m.
- **Hoberg–Rothvoß (SODA 2017), classification: BACKGROUND.** They obtain an $O(\log \mathrm{OPT})$ additive integrality gap for classical bin packing. I investigated whether this could sharpen the large-m upper bound of Chen–Tao–Verschae, but I did not find a rigorous transfer to their convex-cost configuration-LP framework.

I did not locate, among the sources I searched, a later paper claiming a complete resolution of the regime $m = \Theta((1/\varepsilon)^{\theta})$, $\theta \in (1/2, 1]$; the 2025 Mathematics of Operations Research paper appears to be the journal version of the 2022 result above.

Proof sketch

The known ETH lower bound lives at machine count $m_0 = \Theta(\eta^{-1/2})$, where η is the approximation parameter of the *source* instance. My goal is to transfer that hardness to a larger machine count $m = \Theta(\varepsilon^{-\theta})$ for a *target* parameter ε.

The key gadget is extremely simple once one normalizes correctly. Given a hard source instance on m_0 machines, I first scale all its jobs so that the total processing volume becomes exactly m_0; thus the average source load is 1. Then I add

$$M := m - m_0$$

new *anchor jobs*, each of size 2.

Why size 2? Because for the convex cost $x \mapsto x^q$, placing extra load on top of a machine that already carries an anchor of size 2 is always more expensive than placing that same extra load onto an anchor-free machine of load at most 1. Since the average source load is 1, some anchor-free machine always has load at most 1. This lets me *canonicalize* any schedule: repeatedly split multiple anchors apart, then push all source load off anchor machines, never increasing the objective. In the canonical form, each anchor occupies its own machine and all source jobs sit on the remaining m_0 machines.

So the augmented instance cleanly decomposes:

$$\mathrm{OPT}_{\text{aug}} = M \cdot 2^q + \mathrm{OPT}_{\text{source-scaled}}.$$

Hence any $(1+\varepsilon)$-approximation for the augmented instance yields a source schedule with factor

$$1 + \varepsilon + O\left(\varepsilon \frac{M}{m_0}\right).$$

Now $M = \Theta(\varepsilon^{-\theta})$, while $m_0 = \Theta(\eta^{-1/2})$. Choosing

$$\eta \asymp \varepsilon^{2-2\theta}$$

makes the transferred error exactly $O(\eta)$. Therefore, a too-fast PTAS in the target regime would imply a too-fast PTAS for the already-hard source regime $m_0 = \Theta(\eta^{-1/2})$, contradicting Chen–Tao–Verschae.

Conceptually, this yields a hardness exponent that interpolates as

$$\sqrt{1/\eta} \quad \rightsquigarrow \quad (1/\varepsilon)^{1-\theta}.$$

Main result

Theorem 1. *Fix $q > 1$ and $\theta \in (1/2, 1)$. Assume ETH. Then for every $\gamma > 0$, there is no PTAS for $P|||\sum_{i=1}^{m} C_i^q$ on instances satisfying*

$$m = \Theta\big((1/\varepsilon)^{\theta}\big)$$

whose running time is

$$2^{O\left((1/\varepsilon)^{1-\theta-\gamma}\right)} + n^{O(1)}.$$

Equivalently, in this regime every PTAS requires running time

$$2^{\Omega\left((1/\varepsilon)^{1-\theta-o(1)}\right)} + n^{O(1)}.$$

Proof

I first isolate the canonicalization gadget.

Lemma 1 (Canonicalization by anchors)**.** *Let $\widehat{I}$ be any instance of $P|||\sum_i C_i^q$ on m_0 machines whose total processing volume is exactly m_0. Let $m = m_0 + M$, and form an augmented instance I^+ by adding M anchor jobs, each of processing time 2.*

Then every schedule σ for I^+ can be transformed in polynomial time into a schedule σ^{can} such that:

1. *each anchor job occupies its own machine,*
2. *no source job shares a machine with an anchor,*
3. *$F(\sigma^{\text{can}}) \le F(\sigma)$.*

Proof. For each machine i, let $a_i \in \mathbb{Z}_{\ge 0}$ be the number of anchors on machine i, and let $x_i \ge 0$ be the total load of source jobs on machine i. Thus the machine load is

$$L_i = 2a_i + x_i.$$

Define, for each fixed $z > 0$,

$$g_z(t) := (t+z)^q - t^q \qquad (t \ge 0).$$

Since $q > 1$,

$$g_z'(t) = q\big((t+z)^{q-1} - t^{q-1}\big) > 0,$$

so g_z is strictly increasing in t.

We use two local operations.

Step 1: split multiple anchors apart. Suppose some machine u has $a_u \ge 2$. Let

$$R := \{i : a_i = 0\}$$

be the set of anchor-free machines. Since at most M machines carry at least one anchor, we have

$$|R| \ge m - M = m_0.$$

The total source load is m_0, so some machine $v \in R$ satisfies

$$x_v \leq \frac{m_0}{|R|} \leq 1.$$

Move one anchor from machine u to machine v. Before the move, the contribution of u, v to the objective is

$$\big((2(a_u - 1) + x_u) + 2\big)^q + x_v^q = g_2\big(2(a_u - 1) + x_u\big) + \big(2(a_u - 1) + x_u\big)^q + x_v^q.$$

After the move, it is

$$\big(2(a_u - 1) + x_u\big)^q + (x_v + 2)^q = \big(2(a_u - 1) + x_u\big)^q + x_v^q + g_2(x_v).$$

Because $a_u \geq 2$, we have $2(a_u - 1) + x_u \geq 2 > x_v$, and since g_2 is increasing,

$$g_2\big(2(a_u - 1) + x_u\big) \geq g_2(x_v).$$

Hence the move does not increase the objective.

Repeating this step, we eventually obtain a schedule in which every machine carries at most one anchor.

Step 2: evacuate source load from anchor machines. Now every machine has $a_i \in \{0, 1\}$. Suppose some machine u has $a_u = 1$ and $x_u > 0$. As above, among anchor-free machines there exists some v with $x_v \leq 1 < 2$. Move *all* source load x_u from machine u to machine v.

Before the move, the contribution of u, v is

$$(2 + x_u)^q + x_v^q = 2^q + x_v^q + g_{x_u}(2).$$

After the move, it is

$$2^q + (x_v + x_u)^q = 2^q + x_v^q + g_{x_u}(x_v).$$

Because $2 > x_v$ and g_{x_u} is increasing,

$$g_{x_u}(2) \geq g_{x_u}(x_v),$$

so again the move does not increase the objective.

Repeating this step terminates with every anchor alone on its own machine and all source jobs on the remaining m_0 machines. The number of moves is polynomial in the number of jobs, so the transformation is polynomial-time. □

Lemma 2 (Approximation transfer)**.** *Let $\widehat{I}$ be any instance on m_0 machines with total processing volume m_0. Let I^+ be the augmented instance obtained by adding M anchor jobs of size 2, on $m = m_0 + M$ machines.*

If I^+ admits a $(1+\varepsilon)$-approximate schedule, then one can obtain from it, in polynomial time, a schedule for $\widehat{I}$ of value at most

$$\left(1 + \varepsilon + 2^q \varepsilon \frac{M}{m_0}\right) \mathrm{OPT}(\widehat{I}).$$

Proof. By the previous lemma, any schedule for I^+ can be canonicalized without increasing its cost. In a canonical schedule the objective decomposes as

$$F = M \cdot 2^q + F_{\mathrm{src}},$$

where F_{src} is the objective of the induced schedule on the m_0 source machines.

The same canonicalization applied to an optimal schedule shows

$$\mathrm{OPT}(I^+) = M \cdot 2^q + \mathrm{OPT}(\widehat{I}).$$

Indeed, the schedule that combines an optimal schedule for $\widehat{I}$ with one anchor per extra machine is feasible, giving the matching upper bound.

Now let σ^+ be a $(1+\varepsilon)$-approximate schedule for I^+, and canonicalize it. If F_{src} denotes the source part of the canonicalized cost, then

$$M \cdot 2^q + F_{\text{src}} \le (1+\varepsilon)\big(M \cdot 2^q + \mathrm{OPT}(\widehat{I})\big).$$

Hence

$$F_{\text{src}} \le (1+\varepsilon)\mathrm{OPT}(\widehat{I}) + \varepsilon M 2^q.$$

Since the total source volume equals m_0, Jensen's inequality yields

$$\mathrm{OPT}(\widehat{I}) \ge m_0 \cdot \left(\frac{m_0}{m_0}\right)^q = m_0.$$

Therefore

$$\varepsilon M 2^q \le 2^q \varepsilon \frac{M}{m_0}\,\mathrm{OPT}(\widehat{I}),$$

and so

$$F_{\text{src}} \le \left(1 + \varepsilon + 2^q \varepsilon \frac{M}{m_0}\right)\mathrm{OPT}(\widehat{I}).$$

The induced assignment of source jobs on the m_0 non-anchor machines is the desired schedule. $\square$

Proof of the theorem. Assume for contradiction that the theorem is false. Then there exist:

- a fixed $\theta \in (1/2, 1)$,
- a fixed $\gamma > 0$,
- constants $c_1, c_2 > 0$,
- and a PTAS $\mathcal{A}$,

such that for every sufficiently small $\varepsilon > 0$, every instance with

$$c_1(1/\varepsilon)^\theta \le m \le c_2(1/\varepsilon)^\theta$$

is solved by $\mathcal{A}$ in time

$$2^{O\left((1/\varepsilon)^{1-\theta-\gamma}\right)} + n^{O(1)}.$$

Without loss of generality, assume $0 < \gamma < 1-\theta$; if $\gamma \ge 1-\theta$, then the assumed running time is even smaller, so the contradiction only becomes easier.

Now invoke the Chen–Tao–Verschae ETH lower bound as a source hardness theorem. Namely: for every fixed $q > 1$ and every $\delta > 0$, assuming ETH, there is no PTAS for $P\|\|\sum_i C_i^q$ with running time

$$2^{O((1/\eta)^{1/2-\delta})} + n^{O(1)},$$

and the hard family produced there satisfies $m_0 = \Theta(\eta^{-1/2})$.

Fix

$$\delta := \frac{\gamma}{2-2\theta} > 0.$$

Let $a_1, a_2 > 0$ be constants such that for infinitely many sufficiently small $\eta > 0$ there is a hard source instance J_η with

$$a_1\eta^{-1/2} \le m_0 \le a_2\eta^{-1/2}.$$

Take such an η. Let S be the total processing volume of J_η, and scale every job by the factor

$$\lambda := \frac{m_0}{S}.$$

Call the scaled instance $\widehat{J}_\eta$. Then $\widehat{J}_\eta$ still has m_0 machines, and its total processing volume is exactly m_0. Since the objective is homogeneous of degree q, scaling preserves approximation ratios.

Choose a constant

$$K \ge \left(\frac{4 \cdot 2^q c_2}{a_1}\right)^2,$$

and define

$$\varepsilon := \left(\frac{\eta}{K}\right)^{1/(2-2\theta)}.$$

Choose

$$m := \left\lceil c_1(1/\varepsilon)^\theta \right\rceil.$$

Because

$$\frac{m_0}{(1/\varepsilon)^\theta} = O\Big(\eta^{-1/2} \cdot \eta^{\theta/(2-2\theta)}\Big) = O\Big(\eta^{(2\theta-1)/(2-2\theta)}\Big) \to 0 \qquad (\eta \to 0),$$

for all sufficiently small η we have $m > m_0$. Also, for sufficiently small η,

$$m \le 2c_1(1/\varepsilon)^\theta,$$

so $m = \Theta((1/\varepsilon)^\theta)$; after adjusting constants, $\mathcal{A}$ applies.

Set

$$M := m - m_0.$$

Construct the augmented target instance I_η^+ from $\widehat{J}_\eta$ by adding M anchor jobs of size 2. Run $\mathcal{A}$ on I_η^+ with parameter ε. By Lemma 2, we obtain in polynomial time a schedule for $\widehat{J}_\eta$ of ratio at most

$$1 + \varepsilon + 2^q\varepsilon\frac{M}{m_0} \le 1 + \varepsilon + 2^q\varepsilon\frac{m}{m_0}.$$

Using $m \le 2c_1(1/\varepsilon)^\theta$ and $m_0 \ge a_1\eta^{-1/2}$, the error term is bounded by

$$\varepsilon + \frac{2^{q+1}c_1}{a_1}\,\varepsilon^{1-\theta}\eta^{1/2}.$$

Since $\eta = K\varepsilon^{2-2\theta}$, the second term equals

$$\frac{2^{q+1}c_1}{a_1} \cdot \frac{\eta}{\sqrt{K}} \le \frac{\eta}{4}$$

by the choice of K. Also $1/(2-2\theta) > 1$, so

$$\varepsilon = (\eta/K)^{1/(2-2\theta)} = o(\eta) \qquad (\eta \to 0),$$

and therefore $\varepsilon \le \eta/4$ for all sufficiently small η. Altogether the returned schedule is a $(1+\eta)$-approximation for $\widehat{J}_\eta$, hence also for J_η.

Now consider the running time. The augmented instance adds only

$$M = O((1/\varepsilon)^\theta) = O\Big((1/\eta)^{\theta/(2-2\theta)}\Big)$$

jobs, so all reduction overheads are polynomial in $1/\eta$ and n. The exponential part becomes

$$2^{O\left((1/\varepsilon)^{1-\theta-\gamma}\right)} = 2^{O\left((1/\eta)^{(1-\theta-\gamma)/(2-2\theta)}\right)} = 2^{O\left((1/\eta)^{1/2-\delta}\right)}.$$

Thus we have produced a PTAS for the Chen–Tao–Verschae hard family with forbidden running time

$$2^{O\left((1/\eta)^{1/2-\delta}\right)} + n^{O(1)},$$

contradicting their ETH lower bound.

This contradiction proves the theorem. □

Gap to the full statement

What remains open is substantial.

- My theorem gives only a *lower bound*, not a matching upper bound.
- For $\theta \in (1/2, 1)$, it leaves a gap between the new lower bound

  $$2^{\Omega((1/\varepsilon)^{1-\theta-o(1)})}$$

  and the best general upper bound

  $$2^{\tilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}.$$

- At the endpoint $\theta = 1$, my lower bound becomes vacuous. This is consistent with the known polynomial-$1/\varepsilon$ PTAS for $m = \Omega((1/\varepsilon)\log^2(1/\varepsilon))$, but does not settle the exact regime $m = \Theta(1/\varepsilon)$.

So the full characterization asked in the prompt remains open, but we now know that throughout $\theta \in (1/2, 1)$ one cannot hope for a PTAS whose exponential dependence is significantly smaller than $2^{(1/\varepsilon)^{1-\theta-o(1)}}$, unless ETH fails.

Self red-team

Counterexample attempts. I tried to break the anchor-canonicalization lemma in the two only plausible ways.

1. *Two anchors on one machine might be beneficial if the anchor-free machines are already heavily loaded.* After normalization the average source load is 1, and the number of anchor-free machines is always at least m_0, so some anchor-free machine has load at most 1. Since anchors have size 2, the marginal cost of placing an extra anchor on an already-anchor machine is larger than placing it on that least-loaded anchor-free machine. No counterexample survived.

2. *Moving a large source block off an anchor machine could increase cost if that block is huge.* For fixed transferred amount $x > 0$, the increment $g_x(t) = (t + x)^q - t^q$ is increasing in t. Because the anchor base load is 2 and the least-loaded anchor-free machine has load at most 1, moving source load away from an anchor machine is always weakly improving, regardless of how large x is.

3. *The anchor baseline $M2^q$ might swamp the source objective so much that the reduction loses the approximation gap.* The transfer lemma quantifies this exactly: the blow-up is $O(\varepsilon M/m_0)$, not $O(\varepsilon M)$, because the normalized source optimum is at least m_0. This is why choosing $\eta \asymp \varepsilon^{2-2\theta}$ works.

Three likely referee objections.

1. *"Why is there always an anchor-free machine of load at most* 1*?"* Because total source volume is m_0, and at every stage the number of anchor-free machines is at least m_0.

2. *"Are you allowed to normalize the source instance by scaling?"* Yes. Multiplying every processing time by λ multiplies every machine load by λ, and hence multiplies the objective by λ^q; approximation ratios are preserved exactly.

3. *"Does the target machine count really lie in the regime* $\Theta((1/\varepsilon)^\theta)$ *after padding?"* Yes. I choose $m = \lceil c_1(1/\varepsilon)^\theta \rceil$, which is by definition in that regime; moreover $m_0 = o((1/\varepsilon)^\theta)$, so $M = m - m_0 > 0$ for sufficiently small parameters.

Boundary checks.

- As $\theta \downarrow 1/2$, the lower-bound exponent $1 - \theta$ tends to $1/2$, recovering the known hard regime.
- As $\theta \uparrow 1$, the exponent tends to 0, so the theorem becomes weak; this matches the fact that the endpoint $\theta = 1$ is still unresolved.
- For $q = 2$, the local moves become especially transparent:

$$(2 + x)^2 + y^2 - \left(2^2 + (x + y)^2\right) = 2x(2 - y) \geq 0$$

  whenever $y \leq 1$. This sanity-check agrees with the general $q > 1$ proof.

Ideas tried

1. **Direct-product / replication lower bound.** Idea: duplicate the known hard instance r times and argue that the optimum scales linearly when both jobs and machines are replicated. Obstruction: I could not prove an exact identity of the form $\mathrm{OPT}_{rm}(rI) = r\,\mathrm{OPT}_m(I)$. A decomposition of a replicated schedule into r layers naturally spreads one layer over up to rm machines, not m, and that kills the argument.

2. **Algorithmic improvement via additive bin-packing advances.** Idea: Chen–Tao–Verschae's large-m algorithm loses $O(\log^2 m)$; perhaps replacing Karmarkar–Karp by Hoberg–Rothvoß could reduce this. Obstruction: their Algorithm AL3 is not plain bin packing; it is a convex-cost configuration-LP recursion with discarded mass and per-round additive loss. I did not find a rigorous way to import the $O(\log \mathrm{OPT})$ bin-count guarantee into that convex-cost setting.

3. **Padding with very large anchor jobs.** Initial idea: reserve the extra machines by adding huge jobs. Initial obstruction: if anchors are chosen larger than the total source volume, the anchor baseline dominates too much and the multiplicative $(1+\varepsilon)$-slack obscures the source gap. Resolution: normalize the source average load to 1 and use anchors of size 2. This is just large enough to force canonical separation by convexity, but small enough that the baseline contributes only an $O(M/m_0)$ relative factor. This refinement is what made the proof work.

Invoked results

- **Chen–Tao–Verschae ETH lower bound for the $\sqrt{1/\varepsilon}$-machine regime.** *Statement used.* For every fixed $q > 1$ and every $\delta > 0$, assuming ETH, there is no PTAS for $P\|\|\sum_i C_i^q$ with running time
$$2^{O((1/\eta)^{1/2-\delta})} + n^{O(1)}.$$
The same paper states that the hard family behind this lower bound has $m = \Theta(\sqrt{1/\eta})$. *Classification.* TOOL. *Citation.* Chen, Tao, Verschae, SODA 2022 / MOR 2025. *Hypothesis audit.*
  - The theorem requires fixed $q > 1$: this matches the present problem.
  - It assumes ETH: this is exactly the ambient hypothesis of my theorem.
  - It is a lower bound against PTAS running time as a function of the approximation parameter: this is exactly how I use it, with source parameter η.
  - The hard family has $m_0 = \Theta(\eta^{-1/2})$: this is the source regime from which I reduce.

Self-confidence

3 — I checked the internal reduction carefully and the anchor-canonicalization argument is quite robust; my main residual uncertainty is only about how best to package the exact external lower-bound statement from Chen–Tao–Verschae, not about the logic of the new reduction itself.